

microcontrollers. The MCF5225x, Freescale's latest MCU, was created by the India team in a record time. This MCU is well-suited for a broad range of industrial networking, building/lighting control and medical applications that require high performance and connectivity options (Ethernet and USB).

Infochips did the product design (for a US customer) of a transcode device used for video streaming application. It involved 16-layer board design with around 4000 components including more than eight high-speed processors.

What advantages does India offer?

Nirav Shah, director of marketing, Infochips, says that for embedded systems players, India offers the advantages of large domestic market size, low product development cost, embedded expertise including IP solutions, and established working models locally and globally.

Large Indian companies are moving towards higher-end activities in the embedded value chain with increased expertise in specific domains. They are metamorphosing from labour cost arbitrage to being true contributors to product innovation. A growing phenomenon is that of companies either investing in new design operations in India or significantly expanding their existing facilities. This will contribute to the further expansion of the talent pool and maturation of skill sets in Indian design companies, which will be a true differentiator in the long run.

At the same time, as the pressure mounts on foreign companies to reduce costs of product development, they are increasingly looking for offshore partners who can provide cost-effective services.

But this is not without challenges

The challenge and opportunity lie in the complexity that is involved in em-

“In addition to just developing a block, the local design team should make it a fully packaged, independently usable block.”

—Poomima Shenoy, president of India Semiconductor Association (ISA)



bedded design. There is an exponential increase in software complexity. In some areas, the code size is doubling every nine months. More than 70 per cent of the development cost for complex systems such as automotive electronics and communication systems is due to software development and test.

In order to exploit the opportunities, we need to make sure that we have the right resource and skill set levels within the embedded space—such as digital/analogue chip design, embedded software and board design.

There is also the question of how much the local ecosystem has evolved over the years to develop embedded products end-to-end. Indian companies need to provide start-to-finish services. We need to match or exceed the pace of developments that are happening in other countries such as China and Taiwan. High investment costs for R&D need to come down. Inadequate support from government for the capital-intensive semiconductor industry is affecting feeder segments like chip design.

Talent crunch. The fact that large MNCs are setting up manufacturing here is a testament to the fact that the domestic opportunity in India is growing and there will be a huge need for talent in the areas of embedded systems design, development and productisation.

Currently, the embedded space is witnessing a gap between the skill set required and the skill set available. No doubt, India is equipped with the required talent at design implementa-

tion and test levels, but the issue is the number of readily employable engineers, rather than the number of engineers. Institutes have been churning out engineers but they lack industry exposure and are not 'design-aware.' Hence there is a need for training and real-life design exposure before they can become productive.

The industry requires expertise in digital, analogue, software and board design. There has to be a talent transformation from traditional PC applications to embedded designs.

And if we wish to move up the value chain, we don't have the required talent at product specification, architecture and productisation levels as yet.

The Indian embedded system community needs to change its mindset from project focus to product focus. Though this is happening, it is too slow and too little. Additional focus is needed.

Solution to talent crunch

The solution to talent crunch lies in industry and academia collaborations—engineering colleges providing students with hands-on experience. Many engineering colleges are not equipped with tool-kits that the students need to be trained on for joining the industry.

An engineering student in India would require a fair amount of training to reach the level of expertise that his counterpart in the West would reach with six months' training. This is because the engineering course structure is many a time not in line with industry